SDG 13 CO2 Emissions Predictor

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Turning Climate Data into Action

The Challenge

Climate change is accelerating, but many nations lack predictive tools to manage emissions effectively. Without knowing the key drivers of emissions, climate policies are often shots in the dark.

Key Statistics

- **Global Emissions:** ~36 Gt CO_2/year
- **Paris Agreement Goal:** Limit warming to 1.5°C
- **The Need:** Data-driven policy tools for targeted interventions

Visual Placeholder

Rising CO2 Emissions Graph & Climate Impact Imagery

A Random Forest Predictor

I used a **Random Forest Regressor** to predict CO_2 emissions per capita based on six key features from World Bank and UN datasets for 150+ countries.

Model Features

- GDP per Capita
- Energy Use per Capita
- Urban Population (%)
- Industrial Value (% of GDP)
- Renewable Energy (%)
- Total Population

Technical Highlights

- **Algorithm:** Random Forest (outperformed Linear Regression & Gradient Boosting)
- **Performance:** 94% R^2 score, 0.34 MAE

What Drives Emissions?

Key Findings

- **GDP per capita** is the strongest predictor (35% importance).
- **Energy use** is the second strongest (28% importance).
- **Renewable energy %** shows a negative correlation, offering a clear path for intervention.

Demo Insight

”Increasing renewables from 20% to 50%... drops emissions by nearly 2 tons per capita. That’s actionable insight!”

Feature Importance Chart

(Bar chart showing GDP as the highest bar)

Actual vs. Predicted Plot

(Scatter plot showing a tight diagonal line)

Ethical Considerations: Beyond the Metrics

Machine learning isn't neutral. Responsible implementation requires addressing inherent biases.

Identified Biases

1. **Data Availability Bias:** Richer nations have more complete data, potentially leading to more accurate predictions for them.
2. **Historical Bias:** The model learns from the past and may underestimate the impact of aggressive, future-focused renewable transitions.
3. **Economic Bias:** GDP-focused features might miss the nuances of informal economies.

The Core Message

Climate justice means centering affected communities. This tool must be used **WITH** local expertise, not to replace it.

Impact & Recommendations

This model empowers policymakers to:

- ✓ **Prioritize Investments:** Focus on high-impact areas like renewable energy.
- ✓ **Track SDG Progress:** Use annual emissions forecasting to stay on target.
- ✓ **Plan Sustainable Growth:** Find pathways to decouple GDP from carbon emissions.
- ✓ **Allocate Climate Finance:** Make a data-backed case for funding specific interventions.

Example Use Case

A developing nation could use this model to demonstrate to donors that a specific investment in their renewable grid will yield a quantifiable reduction in CO_2 emissions, making the case for climate finance concrete.

Next Steps & Demo

Future Work

- Deploy as a public web app with real-time data integration.
- Expand to include future scenario modeling (”what-if” analysis).
- Partner with climate NGOs to test and refine in real-world policy contexts.

Interactive Web App

Screenshot or Live Demo of a tool like Streamlit or Dash

Thank You

This isn’t just a model—it’s a tool for climate justice. We’re not just coding, we’re coding for a livable planet.

Questions?